On Causal Discovery Methods and Applications to Criminal Justice

CS344 Progress Report

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1 Introduction

This project aims to serve as an empirical field-guide for practitioners seeking to apply causal discovery methods to the problem of reoffending. Work carried out by Niu, Gao, Song, and Li (2024) [24] highlighted the relationship between domain and data characteristics and the performance and efficacy of discovery algorithms. It is therefore important to examine algorithms in a domain specific context.

Concretely, this study benchmarks and explores the properties of a range of causal discovery algorithms on recidivism datasets, compares their relative strengths and weaknesses, and provides a worked example of how one might go about eliciting a causal graph with recidivism as a target outcome. In doing so, it complements existing work such as Shirvaikar and Lakshminarayan's "Rethinking recidivism through a causal lens" (2024) [25], which applied the classical PC algorithm to North Carolina data. By broadening the algorithmic toolkit used and drawing on metrics such as the idea of self-compatibility deriving from Faller et al. (2024) [15], this project hopes to contribute to the corpus of literature by focusing explicitly on the robustness (the extent to which an algorithm is able to weather causal complications and transfer across domains) and stability (the ability of an algorithm to perform on restricted subsets of variables) of several causal discovery algorithms.

The motivation for this area of research is threefold: From a domain perspective, criminal justice is an area where predictive tools are already widely deployed, but where explainability, transparency and the ability to justify decisions are essential, making recidivism an especially natural domain for causal approaches. Second, well constructed causal models are in principle more transportable across jurisdictions and cohorts than because they aim to capture invariant mechanisms. Thirdly, and perhaps most importantly, they provide clear targets for intervention, and thereby highlight vectors through which societies can be protected from the risk of repeat offending to the greatest extent possible, and by which offenders themselves can be diverted from a vicious cycle of lifetime antisocial behaviour. Against this backdrop, this project's specific contribution is to qualify how a range of causal discovery algorithms behave when faced with real-world data and known causal red herrings; The work aims to give practitioners concrete guidance about which methods are credible for recidivism analysis and to understand what their limitations are.

The first term has been spent gaining an understanding of the criminal justice setting and building background knowledge of prevailing academic zeitgeist from multiple perspectives. In addition to this, significant time has been spent digesting the technical prerequisites underpinning causal graphs, examining different ways of addressing mixed type data, and clarifying methodology.

This progress report illuminates work carried out during the first term, evaluates progress against initial objectives (and adjusts these where appropriate), and provides a granular plan for the remaining term.

2 Theoretical Underpinnings of Recidivism

Towards the objective of evaluating causal discovery algorithms in the absence of ground truth, it is imperative that practitioners have a good understanding of the domain, and so are able to differentiate plausible models. Substantial background research has been carried out in the first portion in an effort to elicit domain knowledge supported candidate causal variables. Several approaches were taken: a literature review of criminological perspectives on the theoretical basis of recidivism; a review of inputs to prevalent quantitative risk assessment tools; a review of the winning models in the original NIJ recidivism forecasting challenge and; a review of interventional initiatives and their empirical effect on recidivism. Although there is no space here to expand on all of the above, we present some of the useful contextual knowledge gleaned here. This synthesis of criminological literature and policy evaluations will later serve both as prior knowledge to improve algorithm accuracy where applicable and as a qualitative check for learned graphs.

2.1 Recidivism Literature Review

The literature review consisted of a broad characterisation of criminogenic and non-criminogenic factors as identified in exploratory reading. In this context, and in line with the theory developed by Andrews and Bonta (1990) [7] criminogenic needs simply refer to the subset of dynamic risk factors that "when changed, are associated with changes in the chances of recidivism" (pg. 31 Andrews, Bonta, & Hoge (1990)). Notice that when we read *changed* as *intervened (up)on*, we get a glimpse of a possible causal

structure underlying the process. A summary of the criminogenic factors commonly cited in literature can be found below.

Criminogenic Factors in Literature

Criminogenic Factor	Description	RNR
Anti-social Peer Associations [20, 4, 11, 14]	This encompasses pro-criminal social support and a lack of "anticriminal others" [4, 13].	✓
Anti-social Personality Pattern [20, 4, 11, 13]	As per *The Diagnostic and Statistical Manual of Mental Disorders 5E* [9], this includes traits of but is not limited to: 1. Impairments in self functioning: Encompassing **Identity** and **Self-direction**. 2. Impairments in interpersonal function: Principally issues with **Empathy**, and **Intimacy**. 3. Antagonism: As distinguished by **Manipulativeness**, **Deceitfulness**, **Callousness**, and **Hostility**. 4. Disinhibition: Identified by **Irresponsibility**, **Impulsivity**, and **Risk taking**.	✓
Substance Abuse [20, 2, 4, 22, 11, 14, 13]	Including alcohol and other drugs.	✓
Anti-social Cognition [4, 11, 14, 13]	Anti-social cognition refers to attitudes, thought processes, and beliefs which lead to engagement in anti-social behaviours. For example, anger, defiance, or identification with a criminal identity.	✓
(Lack of) Accommodation [2, 1, 13]	This is inclusive of temporary accommodation and rough sleeping.	
(Lack of) Education / Employment [2, 4, 26, 11, 13]	Identified by poor scores in measures of achievement and fulfilment at school/work.	✓
Familial and/or Marital [4, 11, 13, 8]	Familial and marital bonds contribute in two key ways [4]. They provide a sense of belonging and care, as well as being natural sources of supervision.	✓
Leisure and/or Recreation [4, 11, 8]	Characterised by a lack of engagement or fulfilment derived from non-criminal pastimes.	✓
Finances/Income [18, 21, 19]	This captures financial difficulties.	
Mental Disorders [13, 16, 5, 21]	As distinct from just personality related disorders, including clinical conditions such as schizophrenia.	

The ✓ simply refers to the inclusion of the criminogenic factor in the risk, need, and responsivity model as one of the "Central Eight" - factors which are most strongly associated with recidivism [6].

2.2 Static Risk Factors

Unlike criminogenic risk factors, static risk factors are immutable. This therefore makes them somewhat less interesting for policy-makers. Nonetheless, for completeness we list some common static risk factors

found in literature.

Static Risk Factors affecting Recidivism

Static Risk Factor	Description	RNR
Criminal History [3, 21, 13, 17]	Used to identify a lifespan long pattern of offending.	✓
Age [3, 13, 17, 19]	Recidivism is negatively correlated with age.	
Traumatic Experiences [13]	Adverse psychological or physical trauma experienced in an individual's history.	

2.3 Risk Assessment Tools

Risk assessment is a central part of managing public safety. They are also useful tools insofar as determining the factors deemed relevant to reoffending by prison officers. To that end, the project reviews existing risk assessment instruments and maps their input variables to theory via the "Central Eight" criminogenic needs in order to validate the feature space used in subsequent causal discovery experiments. In brief, the instruments covered were COMPAS, OASys, LSI-R, LS/CMI, OGRS3, RISc, PCRA, ORAS (and all of its constituent variants), HCR-20, Static-99, VRAG, and VRS. This offers a good range of general recidivism tools as well as specialist sexual and violence recidivism tools.

Instruments that are currently in use fall into two categories: actuarial tools and structured professional judgement tools (SPJ), both of which have been shown to outperform [28] unstructured tools common in first generation risk assessment models.

Actuarial instruments such as OASys, COMPAS, and LSI-R score individuals against empirically determined line items. Scores are then translated to a risk as determined by the population of the original development study. These are favoured by public institutions since they are both transparent and fast (and therefore cheap). On the other end of the spectrum, structured professional judgement approaches guide trained professionals through checklists of theoretically motivated factors. The key difference here is that discretion is given to the assessor to interpret how these factors may affect a specific individual (in RNR terms, discretion is given to interpret individual responsivity).

Practically every modern risk assessment instrument is influenced by the risk-need-responsivity framework developed by Bonta, Andrews, and Gendreau [11]. This is most explicit in their use of the Central Eight criminogenic needs, with factors such as accommodation, employment, relationships, associates, and cognitive/attitudinal factors appearing repeatedly and consistently across tools. This convergence seems to provide independent and practice motivated justification for the set of candidate variables identified above, being as the theory aligns with widely used instruments.

At the same time, these tools do not seek to distinguish causal from non-causal predictors and so provide little direction as to which factors make good interventional targets. A causal approach would complement instruments by allowing fewer variables to be focused on and with higher fidelity.

2.4 Rehabilitation Program

Another research angle taken is to consider empirical reviews of rehabilitation programmes and their outcomes. These act as quasi-natural experiments which target and intervene on particular criminogenic needs.

One such example is the Boston Reentry Initiative. Set up in 2001, it was developed to improve chances of success for high-risk young male offenders reintegrating into society. The cohort was specially selected for being on the riskier end of the spectrum and received intensive support, including assignment to a mentor, vocational training, access to housing, employment, and mental health or substance abuse treatment as appropriate. Evaluation showed that the experimental group experienced reductions in rearrests of around 30% over the three-year follow-up period [12]. Although the programme was not randomised and focused on a high-risk violent offending subgroup, its outcomes add credence to the view that factors such as employment, stable accommodation, substance use, and mental health are not only correlated with recidivism but are plausible levers through which reoffending can be reduced.

Another supporting line of evidence came from an evaluation of Ohio's halfway houses (HWHs) and community correctional facilities (CBCFs) which provide structured residential programs to aid

offenders on parole, parole violators serving sanctions, and individuals transferring from custody back into the community. CBCFs in the study generally offered a wide range of services as part of their core offering, including: substance-abuse treatment, employment support, mental-health services, education, financial skills, anger management, and groups split by cognitive ability. In contrast, HWHs saw greater sensitivity to treatment according to risk buckets, with low-risk participants in some halfway houses experiencing no benefit or even negative effects [23], consistent with concerns echoed in wider literature about antisocial peer influence.

Altogether, these results provide quasi-experimental support for several of the variables set forward by risk-need-responsivity theory—particularly employment, education, substance use, mental health, housing, and antisocial associations.

3 Causal Graphs and Algorithm Selection

The second strand of research devoted substantial time to narrowing down the pool of causal discovery algorithms (of which there are many) by determining which might be good candidates for applications within this area. In particular, a challenge that was encountered almost straight away is that of dealing with mixed data. Many current approaches do not inherently offer support for variables of mixed types [27, 29] and so some digging was required to come up with a solution.

There are a few approaches that can be taken to adapting mixed type data for use with single type solutions: (1) discretising continuous data; (2) one hot encoding for categorical variables and; (3) treating discrete data as continuous. However, these hacks suffer from information loss, poor scaling and increasing dimensionality, and model misspecification respectively. Naively conforming variables to the same data type then has unpredictable consequences to the validity of identified structures in an already noisy setting.

Therefore it was decided to focus on applying algorithms with explicit support for heterogeneous variables. Based on the recommendation algorithms set forth by Niu et al. [24], the advice for i.i.d. datasets are split across combinations of linear/non-linear and Gaussian/non-Gaussian according to the characteristics of the dataset. For completeness, we aim to test algorithms across recommendation categories. Unfortunately, since the empirical review did not focus specifically on mixed type algorithms, some recommended algorithms offer no heterogeneous support. We produce a table below adapted from their recommendations, indicating where appropriate support for mixed types.

Niu, Gao, Song, & Li Recommended Algorithms [24]

Category	Recommendation	Mixed-type
linear + Gaussian	GOLEM	
linear + non-Gaussian	DirectLiNGAM	✓
non-linear + Gaussian	GraNDAG NOTEARS CORL	✓
non-linear + non-Gaussian	CORL NOTEARS	✓

LiM extends LiNGAM so is a sibling algorithm to DirectLiNGAM, and DAGSLAM is a direct extension of NOTEARS. The recommended algorithms are therefore well represented in mixed type terms across all but the linear and Gaussian case when considered in the "portfolio" below.

3.1 Selection of Causal Discovery Algorithms

There are a plethora of causal discovery algorithms available which span different methodologies from conditional independence methods, to structural equation model (SEM) based methods, to deep learning based methods. To complicate things further, they make varying levels of assumptions about the underlying structure, including about the distribution of noise and linearity (or non-linearity).

It has been reported that appropriate selection of an algorithm has a significant impact on performance when applied to a particular dataset [24], and it is therefore crucial for the efficacy of this project to select algorithms that may perform well under characteristics of recidivism data, as well as to represent a broad base of algorithms that may indicate which family produce the best results.

1. Linear Mixed Data (LiM)
   Description: LiM is a score based method which assumes an underlying linear SEM and non-Gaussianity to recover causal ordering. The method extends LiNGAM to handle mixed data. Firstly, a global continuous optimisation with acyclicity constraint is used, followed by a combinatorial search.

2. DAGBagM
   Description: DAGBagM is another score-based method that uses composite Bayesian information criterion scores to allow for natural treatment of continuous and binary variables.

3. Hill-climbing (HC)
   Description: Hill-climbing is a simple and commonly used greedy search method whose search space is the DAG space. This provides a useful reference point to test if more complex algorithms make meaningful improvements to performance.

4. Fast Causal Inference (FCI)
   Description: A constraint-based algorithm which can be used with mixed type data via an appropriately mixed type conditional independence test. Importantly, this algorithm does not assume causal sufficiency and therefore allows for unobserved latent confounders. This will provide contrast to algorithms which assume no hidden common causes and will illustrate the sensitivity of inferred recidivism structures to the assumption.

5. Directed Acyclic Graphs Structure Learning via Log-determinant and Augmented Lagrangian for Mixed data (DAGSLAM) [10]
   Description: DAGSLAM is unique here for being representative of a continuous-optimisation algorithm, having been extended from NOTEARS. Linear regression is used for continuous variables where logistic regression is used for categorical variables, combining this into a single score.

6. ReX
   Description: ReX is an experimental explainability-driven approach which uses machine-learning to learn features. SHAP explainers are used to identify candidate parent sets which are then aggregated across bootstrap resamples to produce an adjacency matrix. It is interesting to compare how its behaviour differs to classical constraint and score based methods.

Altogether, this collection has been deliberately constructed to cover distinct paradigms with representation from constraint-based (FCI), score-based (LiM, DAGBagM, Hill-climbing), continuous-optimisation-based (DAGSLAM), and novel hybrid machine learning explanation-driven methods (ReX). In addition, an ML method is considered, alongside conditional independence methods (FCI) and structural equation model based methods (LiM, DAGSLAM).

3.2 Common Properties and Assumptions Made by Causal Discovery Algorithms

Due to constraints imposed by causal identifiability, algorithms are required to make (often strong) assumptions about the data at hand. The three core assumptions are the causal Markov condition, faithfulness, and causal sufficiency (although some algorithms explicitly relax this assumption). The causal Markov condition dictates that in a causal DAG, each variable is independent of all of its non descendants conditioned on its parent set. Faithfulness can be interpreted as saying that any conditional independencies appearing in the data are those implied by the causal Markov condition on the underlying DAG and vice versa. That is, the distribution is faithful to the graphical representation. Finally, causal

sufficiency is the assumption that all of the common causes of every variable in the dataset are also included in the model. That is, there are no latent confounders. In practice, this is difficult to guarantee and often violated. This justifies the inclusion of an algorithm in the benchmarking which relaxes the assumption.

3.3 Performance Metrics

To compare algorithms on synthetic data, the project will use structural Hamming distance (SHD), precision, recall, and the F1 score as primary performance metrics.

SHD quantifies how many edge insertions, deletions, or reversals separate two graphs, and therefore provides an explicit measure of structural accuracy:

$$SHD(\hat{G}, G^*) = \text{ \# edge additions, deletions, and reversals needed to transform } \hat{G} \text{ into } G^* \tag{3.1}$$

Precision and recall represent the proportion of predicted edges that are correct and the proportion of true edges that are recovered, respectively:

$$\text{precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}. \tag{3.2}$$

$$\text{recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}. \tag{3.3}$$

Finally, F1 score includes both aspects in a single metric which penalises false positives and false negatives. That is:

$$\text{F1} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \tag{3.4}$$

For real-world recidivism datasets, where no ground-truth graph is available, the project adopts the self-compatibility framework of Faller et al. [15] as the main diagnostic tool. As proposed by Faller, incompatibility scores are derived from repeated runs on subsets of variables, and act as a proxy for SHD by assessing how stable an algorithm's outputs are under perturbations in the variable set. It is important to note at this stage that self-compatibility is necessary but not sufficient for a good causal model. We therefore use it as a discriminatory measure and place importance on interpreting causal models in the context of qualitative checks against theoretical underpinnings and by examining whether learned graphs align with known criminogenic factors and intervention targets consistent with literature.

4 Other Technical Progress

In addition to the work described above, several other steps have been completed. Metadata tables have been created for the NIJ Georgia dataset, and for two additional datasets with different settings: a dataset capturing recidivism in New York spanning 2008-2020, and a dataset on juvenile recidivism in Florida for 2015-2018. These tables include variable types, coding schema, and definitions where appropriate. This will provide a useful reference during the next phase of the project, which involves the preprocessing of data and setting up the experiment. Preprocessing of the data will involve handling missing entries, duplicates, and any normalisation required.

Alongside the Causal Pitfalls data through which we will test performance under latent confounding and causal paradoxes, these datasets comprise a plethora of tests for the algorithms to undertake, which will allow for a nuanced characterisation of each algorithm's properties.

5 Appraisals, Reflections, and Updated Objectives

Two main unforeseen issues emerged in Term 1: (1) the complexity of mixed-type data handling; and (2) the time required to synthesise criminological frameworks and risk tools as scope creep began to occur. The former led to an explicit focus on algorithms with native mixed-data support (e.g. LiM, DAGBagM, DAGSLAM, FCI with mixed tests), while the latter prompted a conscious decision to narrow the algorithm portfolio to work within the constraints of a third-year project. Despite this phase lasting longer than anticipated, due to the additional clarity, more coherent conceptual framework for selecting variables, algorithms, evaluation metrics, and renewed focus gained during this process, the experimental phase is likely to be positively impacted by taking less time than initially accounted for. Further,

extra time was built into the original Gantt chart in preparation for under or overestimations of time requirements, and so the project is still on track.

Work to date has also improved awareness of how strongly domain-specific assumptions shape both design and interpretation. Many causal discovery algorithms rely on assumptions such as faithfulness and causal sufficiency, which are unlikely to hold fully in criminal-justice data; this underscores the need to combine algorithmic results with domain knowledge, as well as motivating the inclusion of a mixed type algorithm which relaxed the causal sufficiency assumption.

From a project-management perspective, regular supervision meetings and informal check-ins have proved effective in catching conceptual bottlenecks early, such as the challenge of mixed-type data, and in re-aligning objectives without derailing the overall timetable.

Looking ahead, the project will adopt a more focused direction thanks to the crystallised methodology and adopt slightly shorter milestone-driven cycles. For example, aiming to run FCI and LiM on a single CausalPitfalls dataset within a given fortnight to ensure that implementation and experimentation progress in step with ongoing reading and writing.

5.1 Objectives

The objectives broadly remain the same with minor changes in the scope of the causal discovery research and the removal of dataset cataloguing. We describe evidence of achievement for objectives which have been completed and elaborate on those which have been clarified. These changes have been made to reflect the deliberate pivot to a scoped and small but carefully chosen set of algorithms, rather than attempting a broader but invariably less thorough survey.

1. Sourcing Data
   Description: This objective aligns with Objective 1 in the original project specification and was reworked to focus on a couple of datasets in particular, as opposed to a systematic review which would have taken significantly more time. The datasets identified and documented so far (NIJ Georgia, New York, Florida juvenile recidivism, and Causal Pitfalls) satisfy the planned research and scoping phase for this objective by providing diverse synthetic tests and environments for later transportability tests (if time allows).

2. Research into Mixed Type Causal Discovery Algorithms
   Description: This objective aligns with the original Objective 2, "Research into current state-of-the-art causal discovery algorithms", and was refined to focus on mixed type causal discovery algorithms. Several such algorithms are described in the dissertation and catalogued according to their algorithmic properties, fulfilling this objective.

3. Research into Recidivism
   Description: This directly corresponds to Objective 3 in the initial specification and was achieved by taking a multidimensional approach and considering risk assessment tools, literature, as well as quasi-experimental studies.

4. Empirical Comparison of Robustness
   Description: This objective will proceed as planned with a couple of tweaks. Since the Causal Pitfalls dataset provides a thorough and broad range of tests, it may be a case of evaluating which tests are important for capturing recidivism dataset-like performance. If no appropriate dataset exists then more synthetic data can be generated. The metric selection portion of the objective has been achieved as part of the extended research period. This objective involves passing the candidate causal discovery algorithms through a series of "causal challenges" such as Simpson's Paradox.

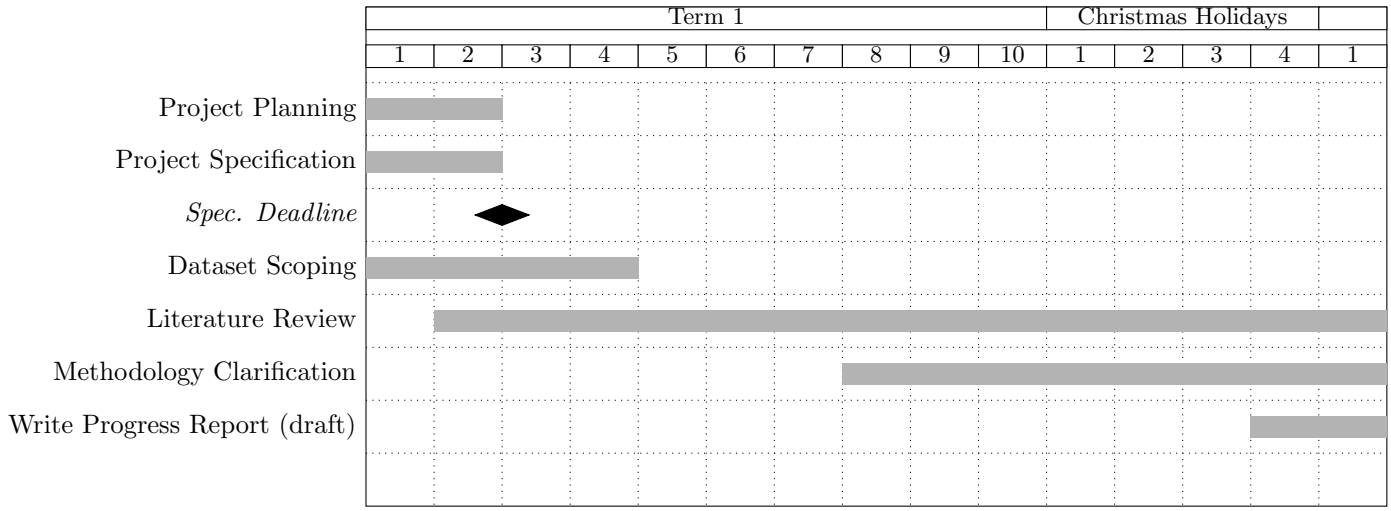
5. Application to Recidivism
   Description: This comprises the largest next step of the project and involves preprocessing data (the creation of tabular metadata undertaken during first term will greatly aid this phase), and running the causal discovery algorithms on this. If time allows, we will examine the ability of elicited models to generalise to other settings by running the same algorithms on the same problem area within a different setting.

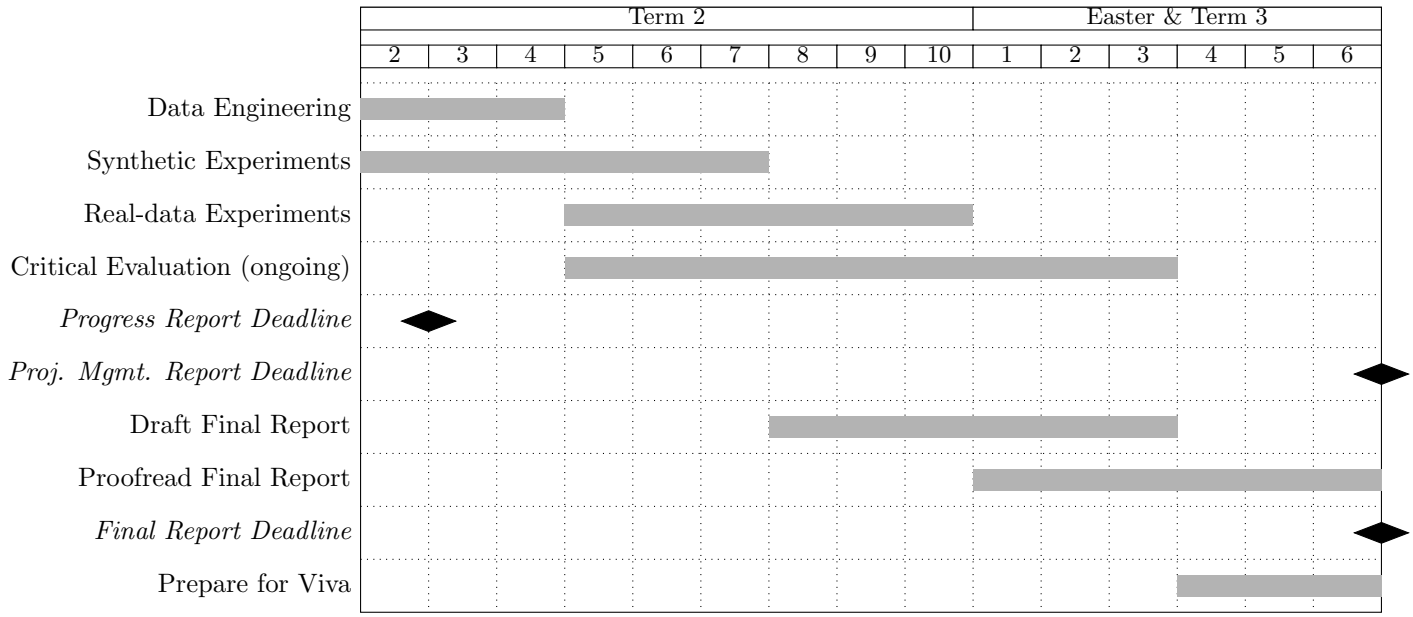
6. Critical Evaluation
   Description: This is an ongoing objective running the length of the project. It is still the goal

to characterise the performance of algorithms under experimental scenarios and real world data conditions to identify the strengths and limitations of approaches. If time allows, a discussion on transportability will be given. Potential improvements to methodology and directions for future work will also be proposed as a part of this objective.

6 Plan

Below is a revised Gantt chart to reflect the plan and workload going forward as well as the actualised workflow of the past few months.





References

[1] Accommodation, homelessness and reoffending of prisoners: Results from the surveying prisoner crime reduction (spcr) survey, 2012. Research summary, applies to England, Northern Ireland and Wales.

[2] Criminogenic needs of prolific offenders, 2023. Better Outcomes through Linked Data (BOLD) programme, Official Statistics.

[3] Reducing reoffending evidence synthesis, 2023. Official report.

[4] D. Andrews, J. Bonta, and J. Wormith. The recent past and near future of risk and/or need assessment. *Crime & Delinquency - CRIME DELINQUEN*, 52:7–27, 01 2006.

[5] D. A. Andrews and J. Bonta. *The level of service inventory-revised*. Multi-Health Systems Toronto, 2000.

[6] D. A. Andrews and J. Bonta. *The Psychology of Criminal Conduct*. Anderson / Routledge, New Providence, NJ, 5 edition, 2010.

[7] D. A. Andrews, J. Bonta, and R. D. Hoge. Classification for effective rehabilitation: Rediscovering psychology. *Criminal Justice and Behavior*, 17(1):19–52, 1990.

[8] D. A. Andrews, J. L. Bonta, and J. S. Wormith. Level of service/case management inventory (ls/cmi) blank form. PDF form, 2004. Reproduction of the LS/CMI instrument.

[9] A. P. Association. *Diagnostic and Statistical Manual of Mental Disorders*. American Psychiatric Association, 2013. See antisocial personality disorder, pp. 659–663.

[10] K. Bello, B. Aragam, and P. Ravikumar. DAGMA: Learning DAGs via M-matrices and a Log-Determinant Acyclicity Characterization. In *Advances in Neural Information Processing Systems*, 2022.

[11] J. Bonta and D. Andrews. Risk-need-responsivity model for offender assessment and rehabilitation. *Rehabilitation*, 6:1–22, 01 2007.

[12] A. A. Braga, A. M. Piehl, and D. M. Hureau. Controlling violent offenders released to the community: An evaluation of the boston reentry initiative. *Journal of Research in Crime and Delinquency*, 46:411 – 436, 2009.

[13] K. S. Douglas, S. D. Hart, C. D. Webster, H. Belfrage, L. S. Guy, and C. M. Wilson. Historical-clinical-risk management-20, version 3 (hcr-20v3): Development and overview. *International Journal of Forensic Mental Health*, 13(2):93–108, 2014.

[14] K. S. Douglas and J. L. Skeem. Violence risk assessment: Getting specific about being dynamic. *Psychology, Public Policy, and Law*, 11(3):347–383, 2005.

[15] P. M. Faller, L. C. Vankadara, A. A. Mastakouri, F. Locatello, and D. Janzing. Self-compatibility: Evaluating causal discovery without ground truth, 2024.

[16] N. I. for Public Management. Compas [computer software], 1996.

[17] A. J. R. Harris, A. Phenix, R. K. Hanson, and D. Thornton. *Static-99: Coding Rules Revised 2003*. Solicitor General Canada, Ottawa, Ontario, 2003.

[18] P. D. Howard. The offender assessment system: an evaluation of the second pilot. 2006.

[19] J. L. Johnson, C. T. Lowenkamp, S. W. Vanbenschoten, and C. R. Robinson. The construction and validation of the federal post conviction risk assessment (pcra). *Federal Probation*, 75(2):16–29, September 2011. NCJ Number 236815.

[20] E. J. Latessa and C. Lowenkamp. What works in reducing recidivism? *University of St. Thomas Law Journal*, 3:521, 2006. University of St. Thomas Law Journal, Volume 3, 2006.

[21] E. J. Latessa, P. Smith, R. Lemke, M. Makarios, and C. Lowenkamp. Creation and validation of the ohio risk assessment system: Final report, July 2009. Funded by the Ohio Department of Rehabilitation and Correction.

[22] N. T. Lokdam et al. Drug use and re-imprisonment: A prospective study of the norwegian offender mental health and addiction (norma) cohort. *Drug and Alcohol Dependence Reports*, 5:100127, dec 2022.

[23] C. T. Lowenkamp and E. J. Latessa. Evaluation of ohio's community-based correctional facilities and halfway house programs, 2002.

[24] W. Niu, Z. Gao, L. Song, and L. Li. Comprehensive review and empirical evaluation of causal discovery algorithms for numerical data, 2024.

[25] V. Shirvaikar and C. Lakshminarayan. Rethinking recidivism through a causal lens, 2024.

[26] T. Skardhamar and K. Telle. Post-release employment and recidivism in norway. *Journal of Quantitative Criminology*, 28, 12 2012.

[27] M. Tsagris, G. Borboudakis, V. Lagani, and I. Tsamardinos. Constraint-based causal discovery with mixed data. *International Journal of Data Science and Analytics*, 6, 08 2018.

[28] J. L. Viljoen, I. Goossens, S. Monjazeb, D. M. Cochrane, L. M. Vargen, M. R. Jonnson, A. J. E. Blanchard, S. M. Y. Li, and J. R. Jackson. Are risk assessment tools more accurate than unstructured judgments in predicting violent, any, and sexual offending? a meta-analysis of direct comparison studies. *Behavioral Sciences & the Law*, 43(1):75–113, 2025.

[29] Y. Zeng, S. Shimizu, H. Matsui, and F. Sun. Causal discovery for linear mixed data. In B. Schölkopf, C. Uhler, and K. Zhang, editors, *Proceedings of the First Conference on Causal Learning and Reasoning*, volume 177 of *Proceedings of Machine Learning Research*, pages 994–1009. PMLR, 11–13 Apr 2022.

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1 Abstract

The problem of uncovering the cause and effect of interacting variables has been of interest to scientists and philosophers for centuries. In particular, the task of separating statistical association with true causal influence from purely observational data is a valuable endeavour with applications to a wide variety of fields - allowing practitioners to employ causal techniques not only to pose counterfactual questions ("what would have happened if A had been B?") but also to anticipate the effects of interventions ("what will happen if we implement this new policy?"). While several surveys and reviews exist within the body of causal discovery literature, many focus only on charting the topology of existing algorithms, evaluation metrics, and datasets, with limited time given to experimental results [1, 2]. And, although recent work such as Niu et al.(2024) [3] conducted a thorough empirical review of 29 algorithms on broad synthetic and real-world benchmarks, performance under known areas of causal difficulty remain under-explored.

This study bridges this gap by empirically evaluating state-of-the-art causal discovery algorithms under targeted scenarios to probe algorithmic robustness and expose weaknesses for each approach.

We will then apply these techniques in direct collaboration with Unilink - a leader in justice technology - to assess state-of-the-art causal discovery algorithms. We leverage Unilink's domain expertise and knowledge from the newly established ML research centre to explore algorithmic robustness on real-world recidivism data absent causal ground truth, applying the self-compatibility concept introduced in Faller et al.[4] as a discriminatory unsuitability measure. By including recent developments in causal discovery algorithms such as ProDAG[5], we incorporate recent Bayesian uncertainty approaches, offering practitioners an updated picture of the capabilities and limitations of modern causal algorithms.

2 Problem Statement & Motivation

There are many benefits to reasoning about the causal structure underlying a process, rather than seeking only to capture associational patterns. For one, the ability to capture the essence of some population independent invariant mechanism allows us to port a causal model between datasets stably, and account for extraneous noise. This is in stark contrast to classical machine learning models which seeks to fit a function to some data. These are often brittle with respect to changes in distributions and suffer from a lack of transportability - the ability of the model to generalise to disparate settings. Secondly, as expounded in Judea Pearl's "The Book of Why" [6], causal understanding allows us to make counterfactual and interventional queries. In practice, this could translate to more flexible and adaptable models, and is of particular interest to policy makers. Due these useful properties, causality has become widely used in fields such as healthcare, political science [7], and economics. In fact, the 2021 Nobel Prize in Economics was awarded to Joshua D. Angrist [8, 9] and Guido W. Imbens [9, 10, 11] "for their methodological contributions to the analysis of causal relationships" [12].

In classical science, causal inference has relied on direct interventions — manipulating variables of interest while holding others fixed — to deduce causal direction. In practice, however, there are many legal, ethical, and logistical reasons why certain experiments are infeasible (the relationship between smoking and lung cancer being an intuitive example). It is for this reason that the problem of recovering causal mechanisms from purely observational data is of interest to researchers. The subfield described above is known as causal discovery, and many algorithms have been developed over the past decades to this end.

One pressing application lies in the area of criminal justice. Explainable intelligence, transparency, and fairness are crucial if such models are to be used in a field that demands accountability and high levels of both legal and ethical scrutiny. In particular, we are interested in the probability of recidivism and in targeting interventions which may succeed in improving rehabilitation and thereby reduce recidivism. The benefits this would provide to society are clear. In Q3 of 2023, the overall proven reoffending rate in the UK was 28.0%, increasing to 31.8% among juveniles, and 62.4% amongst those released from custodial sentences of less than 12 months [13].

The challenge in applying causal discovery techniques is one of prudent algorithm selection, since the choice of algorithm combined with dataset characteristics can lead to results of variable quality [3]. To this end, we aim to empirically evaluate a number of algorithms for suitability of use with recidivism data. By working in collaboration with Unilink, we gain access to a rich testbed, improving the reliability of our results, and better aligning our experimental methods to reflect industry needs.

3 Objectives

1. Sourcing Data
   Description: This will involve a systematic review of available datasets as well as potentially the creation of several custom datasets. The datasets will then be evaluated to identify which are most suitable to test against. Careful selection of benchmarks here will allow us to stress algorithms in specific ways which illuminate algorithmic behaviour under challenging conditions such as selection bias and confounding.

2. Research into Current State-of-the-art Casual Discovery Algorithms
   Description: This includes performing a literature review of peer-reviewed publications over the last year (since the work of Niu et al.), summarising key assumptions and theoretical foundations of each algorithm, and the creation of a summary table with key features. This will allow us to provide the most complete picture of the state of causal discovery algorithms.

3. Research into Recidivism
   Description: This provides more background to the data we will be evaluating algorithms on, and allows us to synthesise expert hypotheses which could be used to recover partial causal models that may be used to evaluate our algorithms (to an extent).

4. Empirical Comparison of Robustness
   Description: This objective involves designing experiments to test algorithms under conditions challenging for causal discovery. The appropriate metrics to quantify performance such as structural hamming distance will also be chosen, catalogued, and computed for each experiment. The results of these experiments can then be visualised.

5. Application to Recidivism
   Description: Using the concept of self-compatibility to determine suitability, we choose candidate algorithms and apply causal discovery techniques to discover potential causal relationships influencing recidivism in the target dataset. The results can then be interpreted and visualised, highlighting any implications.

6. Critical Evaluation
   Description: This involves characterising the performance of algorithms under the experimental scenarios, identifying strengths and limitations of each approach. A discussion of reliability and transportability of findings will be given, along with potential improvements and directions for future work.

4 Methodology

4.1 Methods of Sourcing Data

We can broadly separate the datasets used into two groups: ones used to isolate and evaluate algorithmic robustness, and the target recidivism datasets.

We discuss below the datasets used for testing:

1. CausalPitfalls [14]
   Description: A data suite containing 75 CSV files originally intended to test LLM models under classical causal inference fallacies.

2. CausalBench [15]
   Description: A large dataset consisting interventional data from single-cell perturbation experiments. This will be useful in testing algorithms across interventional data.

3. Custom Synthetic Data
   Description: In cases where particular causally challenging scenarios are not covered in existing datasets, we construct ground truths with the desired properties and sample from the DAG. With regards to noise, we generate both linear and non-linear noise samples and examine robustness under increasing noise, as well as varying the distribution of noise.

4. NIJ Georgia Department of Community Supervision Recidivism Data [16]
   Description: Data on 26,000 individuals under post-incarceration supervision between January 1 2013, and December 31 2015. The data is provided by the Georgia Department of Community Supervision.

5. Data provided by Unilink
   Description: There may be the option to run algorithms on Unilink data. This may have to be done by the Unilink team for reasons of privacy, but we may nonetheless obtain informative results for the purposes of our study.

6. Other Recidivism Datasets
   Description: We aim to identify at least one recidivism dataset from a different region so as to examine the transportability of our recovered causal models.

4.2 Methods of Experimentation

We test algorithms on both real world data as well as synthetic data generated by ground truths of increasing variables, which we sample with increasing amounts of noise. We also vary the distribution of noise, extending beyond Gaussian noise to heavy-tailed distributions, for example. This characterises the ability of algorithms to cope with challenging conditions where one or more assumptions may be broken.

Once we have identified suitable algorithms, we run these on our recidivism data, recovering hypothetical underlying causal structures. These can be compared across different environmental settings to qualify the transportability and robustness of the derived models.

4.3 Methods of Evaluation

The performance of causal discovery algorithms on synthetic datasets can easily be compared to the known ground truth. We will use various metrics such as structural hamming distance between the learned DAG and the ground truth to quantify recovery accuracy.

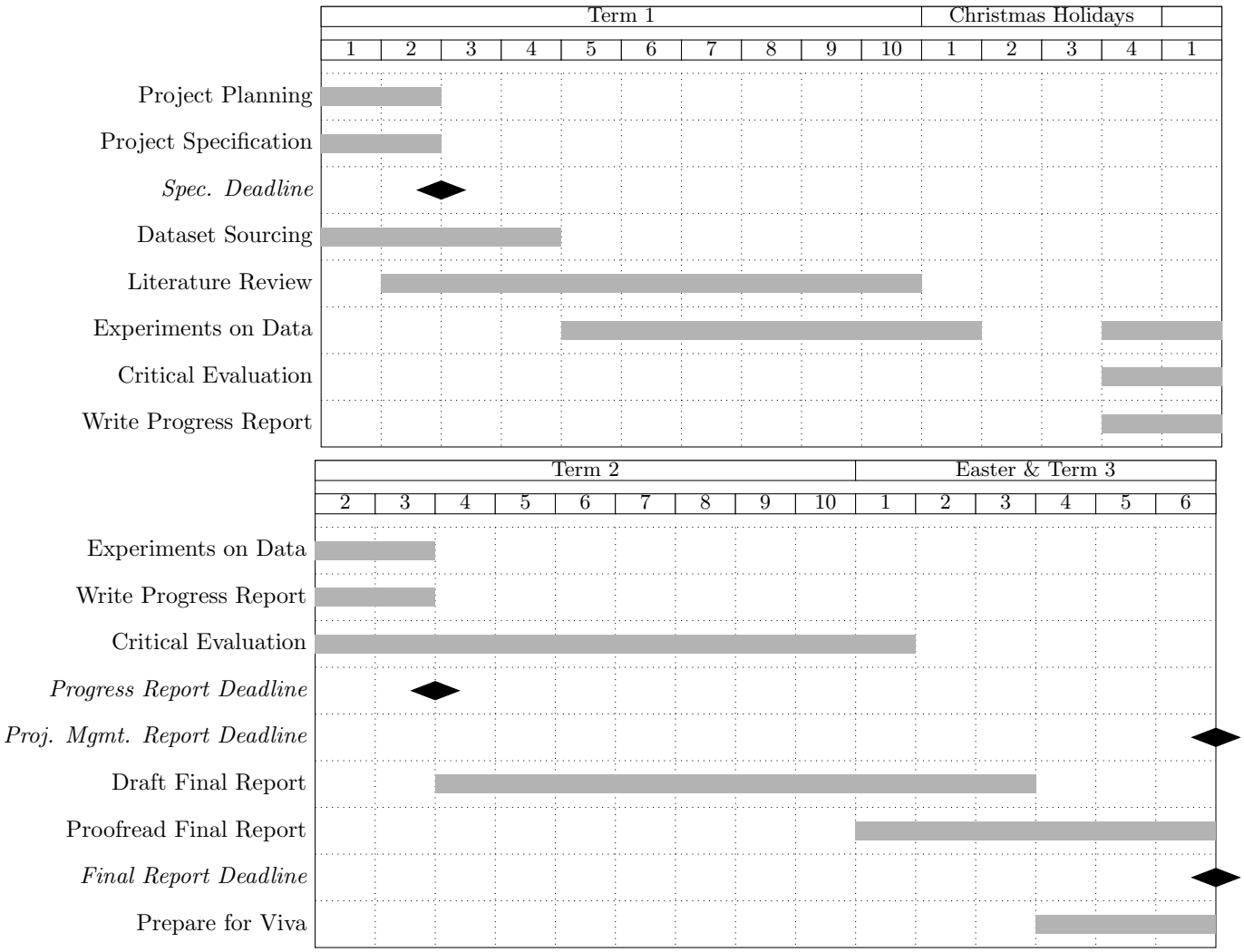
In contrast, evaluating the performance of causal discovery algorithms on real data is challenging due to the absence of causal ground truths accompanying these datasets. We tackle this challenge by applying the self-compatibility technique outlined in the paper "Self-Compatibility: Evaluating Causal Discovery without Ground Truth" [4], and use the proposed incompatibility score as a proxy for structural hamming distance. Abstractly, the incompatibility score is computed by examining the "stability across subsets of variables" [4] of a causal discovery algorithm. It should be noted that while self-compatibility is a necessary criterion for a good causal model, it is not sufficient on its own. We therefore focus on the implication that certain models are not suited for a given dataset.

As a further exploration of robustness, we may also test datasets via partitioning, transportability between different settings, and compatibility with expert hypotheses as described in recidivism literature.

Finally, we also hope to use a traditional machine learning model to illuminate the benefits in predictive performance (if any) of causal methods over classical methods.

5 Timetable

Below is a Gantt Chart depicting a working schedule for the project. Notably, we take into account the internship I will need to work around during Christmas Holidays as well as a block of time completely off.



6 Resources & Risks

6.1 Resources

The following is an indicative list of resources required to carry out the project.

1. Git along with online interface GitHub.
   Git will be used as version control for the project, as well as to store backups of the codebase remotely. GitHub provides a simple user friendly interface to use with Git and is thus the platform of choice.

2. Python
   The primary language used to test algorithms thanks to libraries such as NetworkX which simplifies working with DAGs. Many of the original papers feature an accompanying repository containing a Python implementation, and we will use such implementations where available.

3. Julia
   The ProDAG implementation utilises Julia. Conveniently, Google's Colab offers native support for this, reducing the need for the use of multiple platforms.

4. Google Colab
   Google's Colab platform provides access to cloud compute resources with which we will run causal discovery algorithms.

5. Google Drive
   A cloud storage platform used in conjunction with Colab.

6. Overleaf
   The LaTeX editor of choice.

7. Personal laptop
   Predominantly used as a thin client, with heavy computational tasks delegated to cloud computing resources via Google Colab or alternatively, the department's batch compute resources.

6.2 Risks

We detail below some potential risk vectors associated with the project as well as mitigating points for each.

1. Failure of personal equipment
   Mitigated by frequent backups via GitHub and personal Google Drive facilities. Decentralised storage reduces single point of error and allows easy access from other devices where work can be picked up where it was left.

2. Human error
   Mitigated by prudent use of version control and frequent commits (once per development session as a best practice) so that a granular and complete branch exists over the code's lifecycle. In this way, any code breaking changes can be walked back without much impedence.

3. Cloud service outages
   As mentioned above, using GitHub alongside Google Drive reduces the dependency on one service provider and builds robustness into the development process. Further, in case of prolonged outage which may affect the availability of compute resources, the department's batch compute system can be used instead.

4. Injury/Illness
   This has been mitigated within the project timetabling by using conservative estimates of time required for each phase, along with dedicated time off during the Christmas holidays.

5. Algorithms fail to produce substantive results
   It is plausible that due to the restrictive assumptions required by certain algorithms, the results produced will be not be credible. In that case, it will be sufficient and valuable in its own right to document these shortcomings and incorporate them into the critical evaluation of the algorithm.

7 Legal, Social, Ethical & Professional Considerations

The project does not pose significant legal, social, ethical, or professional considerations.

References

[1] Matthew J. Vowels, Necati Cihan Camgoz, and Richard Bowden. *D'ya like DAGs? A Survey on Structure Learning and Causal Discovery.* 2021. arXiv: `2103.02582 [cs.LG]`. URL: `https://arxiv.org/abs/2103.02582`.

[2] Neville K. Kitson et al. *A survey of Bayesian Network structure learning*. 2022. arXiv: 2109.11415 [cs.LG]. URL: https://arxiv.org/abs/2109.11415.

[3] Wenjin Niu et al. *Comprehensive Review and Empirical Evaluation of Causal Discovery Algorithms for Numerical Data*. 2024. arXiv: 2407.13054 [cs.AI]. URL: https://arxiv.org/abs/2407.13054.

[4] Philipp M. Faller et al. *Self-Compatibility: Evaluating Causal Discovery without Ground Truth*. 2024. arXiv: 2307.09552 [cs.LG]. URL: https://arxiv.org/abs/2307.09552.

[5] Ryan Thompson, Edwin V. Bonilla, and Robert Kohn. *ProDAG: Projected Variational Inference for Directed Acyclic Graphs*. 2025. arXiv: 2405.15167 [stat.ML]. URL: https://arxiv.org/abs/2405.15167.

[6] Judea Pearl and Dana Mackenzie. *The Book of Why: The New Science of Cause and Effect*. 1st. USA: Basic Books, Inc., 2018. ISBN: 046509760X.

[7] Luke Keele. "The Statistics of Causal Inference: A View from Political Methodology: Fig. 1". In: *Political Analysis* 23 (July 2015). DOI: 10.1093/pan/mpv007.

[8] Joshua Angrist, Victor Lavy, and Analia Schlosser. "Multiple Experiments for the Causal Link between the Quantity and Quality of Children". In: *Journal of Labor Economics* (Oct. 2010).

[9] Susan Athey and Guido W. Imbens. "The State of Applied Econometrics: Causality and Policy Evaluation". In: *Journal of Economic Perspectives* 31 (2 May 2017), pp. 3–32.

[10] Guido W. Imbens. "Potential Outcome and Directed Acyclic Graph Approaches to Causality: Relevance for Empirical Practice in Economics". In: *Journal of Economic Literature* 58 (4 Dec. 2020), pp. 1129–1179.

[11] Guido W. Imbens. "Causality in Econometrics: Choice vs Chance". In: *Econometrica* 90 (6 Nov. 2022), pp. 2541–2566.

[12] *The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2021 - Press Release*. Accessed: 2025-10-08. Royal Swedish Academy of Sciences. Oct. 11, 2021. URL: https://www.nobelprize.org/prizes/economic-sciences/2021/press-release/.

[13] Ministry of Justice. *Proven reoffending statistics: July to September 2023*. Accessed: 2025-10-11. UK Government. 2023. URL: https://www.gov.uk/government/statistics/proven-reoffending-statistics-july-and-september-2023/proven-reoffending-statistics-july-to-september-2023.

[14] Jin Du et al. *Ice Cream Doesn't Cause Drowning: Benchmarking LLMs Against Statistical Pitfalls in Causal Inference*. 2025. arXiv: 2505.13770 [cs.AI]. URL: https://arxiv.org/abs/2505.13770.

[15] Mathieu Chevalley et al. *CausalBench: A Large-scale Benchmark for Network Inference from Single-cell Perturbation Data*. 2023. arXiv: 2210.17283 [cs.LG]. URL: https://arxiv.org/abs/2210.17283.

[16] Office of Justice Programs Department of Justice. *NIJ's Recidivism Challenge Full Dataset*. Metadata updated March 12, 2025. Department of Justice, 2025. URL: https://catalog.data.gov/dataset/nijs-recidivism-challenge-full-dataset.